

Programmable Pan-Tilt Camera Platform

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Stage: Mechanical assembly and motion-control verification

Status: Completed — functional test passed

Objective

The objective of this stage was to design, fabricate, assemble, and test a two-axis camera platform controlled by an Arduino UNO. The platform needed to pan left and right, tilt up and down, return to a known center position, and stop before the brackets reached their mechanical limits.

Design and Fabrication

I designed the pan-tilt bracket assembly in Autodesk Fusion 360 using the camera and mounting-hole measurements collected from the hardware. The design separated the base, pan support, tilt support, camera mount, and electronics mounting surfaces. I fabricated the brackets from aluminum and assembled the structure with machine screws and two micro servos.

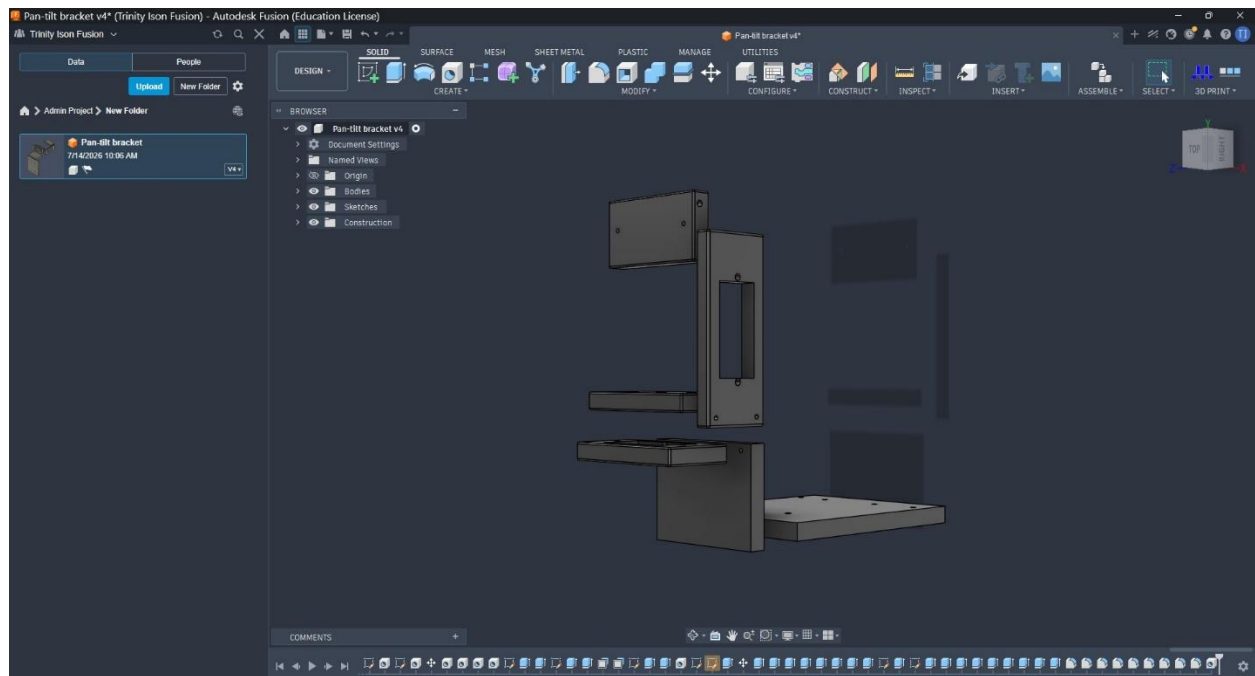


Figure 1. Fusion 360 model showing the custom pan-tilt bracket components and assembly layout.

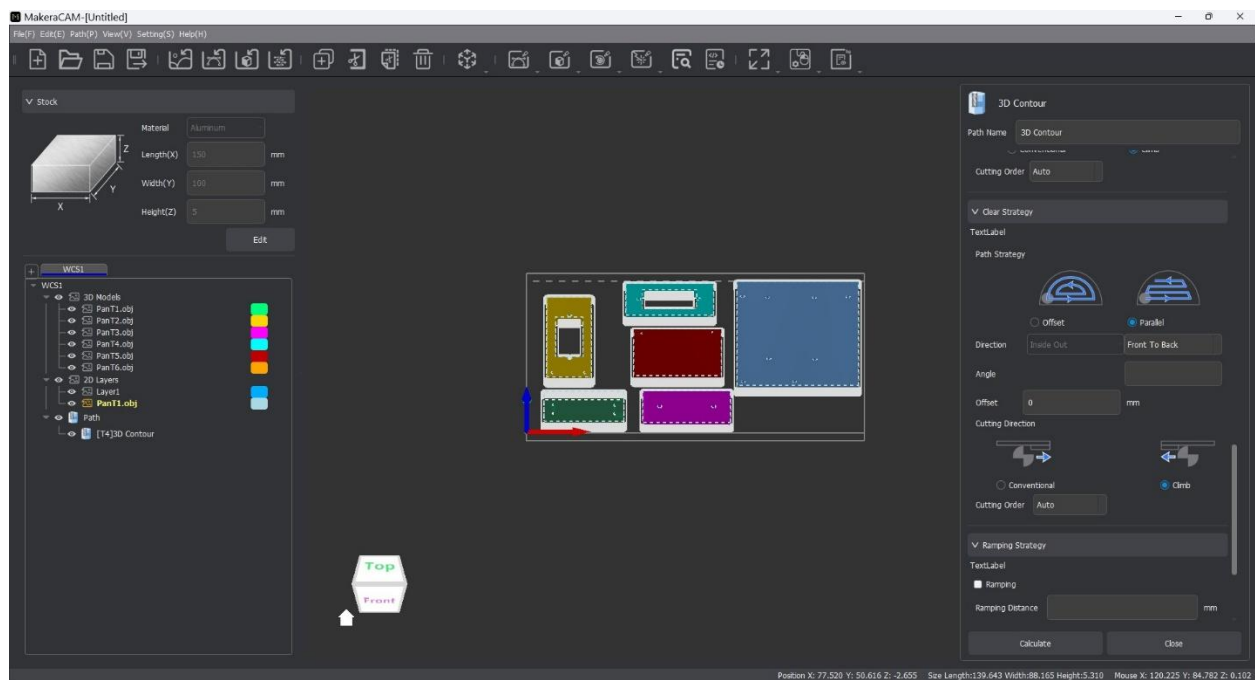


Figure 2. Makera CAM custom tool paths.

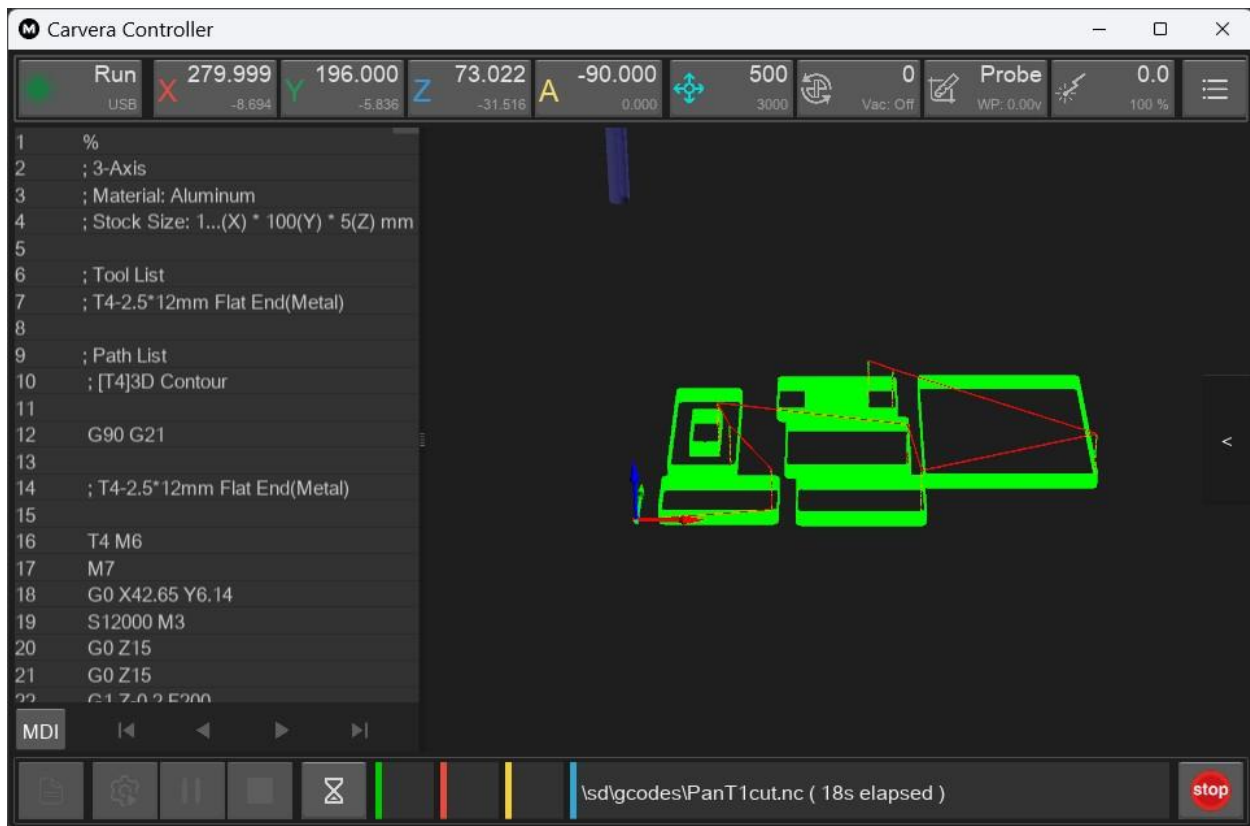


Figure 3. Makera Controller

Hardware and Software

- Arduino UNO-compatible controller with SmartRobot control shield
- Two micro servos for independent pan and tilt movement
- Camera module mounted to the custom aluminum bracket
- Autodesk Fusion 360 for mechanical design
- Arduino IDE 2.3.10 with the Servo library version 1.2.2

Figure 2. Completed camera platform with the Arduino controller, two-axis servo mechanism, and mounted camera.

Control Configuration

Axis	Arduino Pin	Safe Range	Center
Pan	D10 / P10	60 to 120 degrees	90 degrees
Tilt	D11 / P11	85 to 110 degrees	90 degrees

Programming and Test Procedure

I used the Arduino Servo library to attach the pan servo to pin D10 and the tilt servo to pin D11. Both servos were commanded to 90 degrees to establish the center position. I first tested movement in small five-degree steps, then increased the range gradually while watching for mechanical binding, servo buzzing, and cable tension.

1. Centered both servos at 90 degrees and aligned the brackets mechanically.
2. Verified small movement on each side of center before increasing travel.
3. Confirmed that lower tilt values move the camera downward.
4. Set the downward tilt limit at 85 degrees to protect the mechanical stop.
5. Tested pan from 60 to 120 degrees and tilt from 85 to 110 degrees.
6. Returned both axes to 90 degrees and confirmed that the system remained stable.

Problems Encountered and Corrections

- **Upload connection:** The sketch compiled, but the Arduino initially failed to upload because the selected COM port was unreliable and the camera cable was connected to the TXD/RXD serial pins. I moved the USB cable to a different computer port, selected the correct Arduino UNO COM port, and temporarily disconnected the TXD/RXD camera connector during upload.
- **Mechanical alignment:** The first motion test moved the camera to the far-right and downward limits. This showed that the servo horns and brackets needed to be aligned with the electrical 90-degree center. I centered the servos, repositioned the mechanism, and repeated the test using five-degree movements.

- **Unsafe command avoided:** A proposed tilt value of 280 was removed before upload because standard servo angle commands must stay between 0 and 180 degrees. I replaced it with the verified safe tilt range of 85 to 110 degrees.

Final Results

The completed platform moved correctly on both axes. The pan and tilt tests finished without binding, continuous buzzing, or pulling the camera wires. The system returned to the 90-degree center position after each test. The verified limits provide a controlled movement envelope that protects the brackets and servo gears.

Conclusion and Next Step

This stage is complete. The project demonstrates mechanical CAD, fabrication, embedded programming, servo control, calibration, troubleshooting, and software safety limits. I am closing the current project as a Programmable Pan-Tilt Camera Platform. Automatic visual target tracking can be added later as a separate improvement after the camera communication and image-processing stages are developed.

References

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